SoundMatch-SR: The Rendering Transformation Is Learnable, the Taxonomy Is Not—A Dissociation in Synthetic–Real Sound-Effect Retrieval

Elliott Ash
ETH Zürich
ashe@ethz.ch
https://github.com/elliottash/soundmatch-sr

Abstract

Perceptual studies report that listeners cannot reliably distinguish a well-made Foley or text-to-audio rendering of an event from a real recording of the same event [Nordahl et al., 2010]: the *identity* of the event survives the rendering. I ask whether modern audio embeddings represent that identity invariantly to how a sound was produced, and I find a sharp *dissociation*. I introduce **SoundMatch-SR**, a benchmark for cross-domain (synthetic–real) sound-effect retrieval, built on (i) the category-aligned DCASE-2023 Task-7 Foley corpus [Choi et al., 2023] (5,550 real and 25,900 synthetic clips from 37 generator systems, 7 classes) and (ii) a new 34-category benchmark labelled with the Universal Category System (FSD50K [Fonseca et al., 2022] real audio, CLAP-verified, with 10,420 instance-level Stable-Audio-Open [Evans et al., 2024] twins). Off-the-shelf encoders carry a real, linearly-accessible synthetic–real gap (CLAP: 0.161 mAP; Proxy-A-distance 1.27). A lightweight head on frozen embeddings closes it *within* the training taxonomy ($0.161 \rightarrow 0.018$), but a *class-supervised* head does *not* transfer to unseen categories: held-out classes collapse below the frozen baseline, on both 7 and 34 categories. The remedy is to train on the *pairs*, not the labels: an instance-contrastive objective (a clip and its generated twin) learns the rendering *mapping* rather than category identity, and generalizes—on categories never seen in training it retrieves the exact real twin with $\mathrm{R@1} = 0.83 \pm 0.05$ (vs. 0.64 frozen, 0.33 class-supervised), across five folds and three encoders (CLAP, PANNs, AST). Category *clustering* does not transfer for any objective. I release the benchmark, the corpus recipe, the trained heads, and a reusable embedding transform.

1 Introduction

Foley research finds that well-synthesized everyday sounds are perceptually indistinguishable from recordings of the same action; the event's identity survives the synthesis-to-recording boundary for a human listener [Nordahl et al., 2010]. I make this measurable for machine representations:

> *Is "sound identity" recoverable in embedding space, invariant to the rendering process—and if so, does that invariance transfer to sounds the model was not trained on?*

Preprint. Under review.

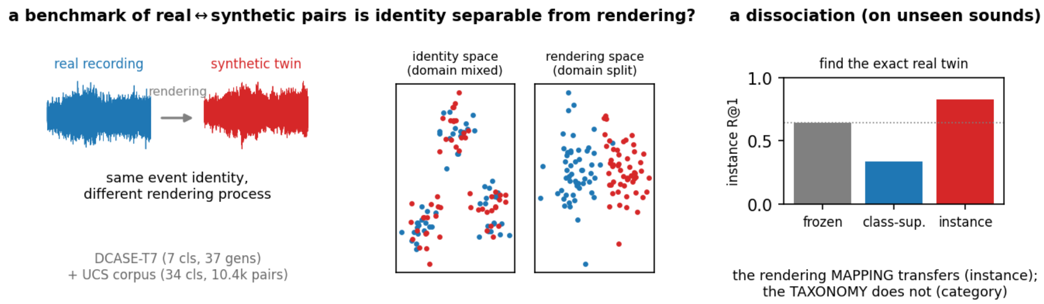


Figure 1: **SoundMatch-SR.** I pair each real recording with a synthetic twin of the same event and ask whether an embedding can represent the event *identity* invariantly to the rendering. The synthetic-to-real *mapping* transfers to sounds never seen in training (instance retrieval, right), while category identity does not.

I operationalize it as retrieval rather than detection: detection rewards *exploiting* the synthetic–real gap, whereas retrieval and invariance reward *removing* it, which is the useful capability for sound-effect search, real/synthetic deduplication, and generative-audio evaluation. My answer is a dissociation that, to my knowledge, is new: the synthetic-to-real *transformation* is a learnable, transferable object, while category identity is not.

Contributions. **(C1)** a two-part benchmark—a controlled 7-class corpus (DCASE-T7) and a diverse, instance-paired 34-category corpus—with leakage-safe splits and a reproducible generation recipe; **(C2)** a measured dissociation: an instance-contrastive objective generalizes the synthetic-to-real mapping to unseen categories, whereas class-supervised invariance does not (and can degrade unseen classes below baseline), robust over five folds and three encoders; **(C3)** a fidelity-spectrum analysis showing the instance head recovers the mapping across twin-fidelity levels, collapsing only when the twin is independent of its source; **(C4)** released artifacts—benchmark, recipe, CLAP-verification, trained heads, and a reusable transform that adjusts any new clip's embedding.

2 Related Work

Audio-representation benchmarks. HEAR [Turian et al., 2022] and related suites evaluate frozen embeddings across classification, retrieval, and clustering, but treat all audio as one domain; none tests invariance to *how* a sound was produced.

Synthetic-audio detection. Deepfake-environmental-audio detectors built on DCASE-T7 pursue the mirror objective (exploit the gap); I remove it, and reproduce detection as a controlled *sensitive* head.

Domain adaptation and invariance. I draw on Proxy-A-distance [Ben-David et al., 2010], domain-adversarial training [Ganin and Lempitsky, 2015], Deep CORAL [Sun and Saenko, 2016], supervised contrastive learning [Khosla et al., 2020], and IRM [Arjovsky et al., 2019]. My generalizing objective is instance-level in the sense of instance discrimination [Wu et al., 2018], and my negative result—that class-level supervision does not transfer the invariance—is a domain-generalization phenomenon.

3 The SoundMatch-SR Benchmark

Core corpus (DCASE-T7). The DCASE-2023 Task-7 corpus [Choi et al., 2023] is a ready-made, category-aligned real/synthetic set across 7 classes: 5,550 real clips (sourced from FSD50K, UrbanSound8K [Salamon et al., 2014], BBC, and Freesound; I use DCASE's source-disjoint dev/eval as train-val/test) and 25,900 synthetic clips from 37 generator systems. It is the controlled setting for a clean gap measurement and a leave-one-class-out probe.

Table 1: The two heads on DCASE-T7 (left) and the bridging ablation (right). The invariant head closes the gap and organises the space by event; the sensitive head is its mirror. Cross-domain supervised contrast does the work.

head	control	cross mAP	sil(ev/dom)	ablation	gap	PAD
frozen	0.935	0.774	0.18 / 0.04	supcon only	$+0.019$	1.38
invariant	0.988	0.970	0.75 / 0.00	+CORAL	$+0.017$	1.38
sensitive	0.206	0.157	-0.04 / 0.75	+DANN	$+0.019$	1.38
				+DANN+IRM	$+0.018$	1.38

UCS corpus. To test generalization I need many categories and instance-level pairs. I label real audio with the Universal Category System (UCS), mapping the 200 AudioSet [Gemmeke et al., 2017] labels of FSD50K onto 34 UCS categories spanning all morphologies (impact, texture, tonal, whoosh, vocal, mechanical, ambience). I *verify* every clip with zero-shot CLAP [Wu et al., 2023] (keep iff its label is top-5 by audio–text cosine), retaining 10,420/13,579 (77%) and removing FSD50K's noisiest multi-labels. For each verified anchor I generate a Stable-Audio-Open *audio-init* twin [Evans et al., 2024] conditioned on the real clip, yielding 10,420 instance pairs. Splits follow FSD50K's train/val/eval, keyed on the Freesound uploader so crops of one recording never straddle the boundary.

Tasks and metrics. *Category retrieval*: query a clip; rank opposite-domain clips; relevant iff same category. I report mAP and a same-domain control, so the headline *domain gap* = control − cross. *Instance retrieval* (UCS): relevant iff the exact cross-domain twin (shared instance id); R@1 and MRR. I report bootstrap CIs, and on the UCS corpus a k-fold leave-classes-out protocol (Sec. 5). Full per-class, per-split statistics and the datasheet are in the appendix.

4 Method

I train a small MLP head (CLAP-$512 \to 256$, ℓ_2-normalised) on *frozen* embeddings; the head is the adjusted embedding and applies to any new clip. The objective decides what it learns. The **invariant** head is class-supervised contrastive with cross-domain positives up-weighted (optionally with DANN/IRM): it pulls same-category clips together and removes domain. The **sensitive** head is domain-supervised within category: the deliberate mirror, organising the space by rendering—its real-vs-synthetic direction is a fidelity axis. The **instance** head uses as positive a clip and *its own* generated twin (with pair-aware batching so twins co-occur in a batch); it learns the synthetic-to-real *mapping* rather than class identity.

5 Experiments

Frozen encoders carry a real gap (Table 2). The synthetic–real axis is real and linearly accessible in both CLAP and PANNs [Kong et al., 2020].

Within-taxonomy the gap closes; the two heads are mirror images. The invariant head closes the DCASE gap $0.161 \to 0.018$ while *raising* same-domain control ($0.935 \to 0.988$) and making the space event-organised (silhouette by event $0.18 \to 0.75$, by domain $0.04 \to 0.00$); the sensitive head flips it to organise by rendering (PAD 1.88, silhouette by domain 0.75). Cross-domain supervised contrast alone closes the gap; DANN/CORAL/IRM add < 0.002 (Table 1); the geometry flip is visible in Fig. 2.

The closed-world failure. Holding one class out of training and evaluating on it, the held-out class collapses *below* frozen (keyboard $0.69 \to 0.43$, gunshot $0.81 \to 0.31$, rain $0.75 \to 0.30$). Adding categories does not fix it (34-class: $0.40 \to 0.19$).

The dissociation (Table 3, Fig. 3a). Training on the pairs, the instance objective generalizes the mapping to unseen categories (instance-R@1 0.83, every fold), class-supervision *hurts* it (0.33), and no objective transfers category clustering (all $\leq$ frozen on unseen-category mAP). It holds on three encoders—CLAP, PANNs, and AST [Gong et al., 2021] (instance R@1 $0.83/0.75/0.95$ vs. class $0.33/0.38/0.38$)—so it is a property of the objective, not of any one encoder.

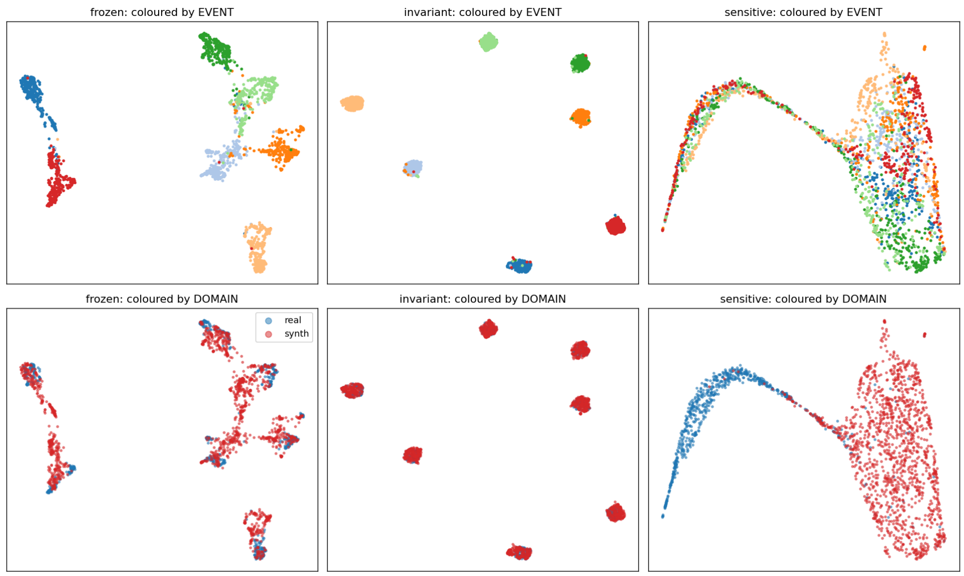


Figure 2: UMAP of the DCASE-T7 test set, frozen vs. the two heads, coloured by event (top) and by domain (bottom). The invariant head tightens event clusters with real/synth overlapping within each; the sensitive head splits real from synth and smears events—the geometry flips.

Table 2: Frozen encoders: synthetic–real gap on DCASE-T7 (test split).

frozen	control	synth→real mAP	gap	PAD	domain-probe
CLAP	0.935	0.774 [0.766,0.783]	+0.161	1.27	0.90
PANNs CNN14	0.798	0.674	+0.124	1.39	0.90

Not near-copies (Fig. 3b). Across twin-fidelity levels (measured as the CLAP cosine of a twin to its source) the instance head holds up (advantage $+0.43..0.52$ over frozen) and collapses to chance only when the twin is independent of its source—so it recovers a rendering mapping, not a near-copy shortcut.

Cross-generator scope. Adding 670 ElevenLabs (text-only) twins captioned from each clip's metadata, instance correspondence is weak (frozen R@1 0.11 vs. 0.55–0.98 for audio-init), because the caption is a bottleneck: the twins are category-appropriate but not renderings of the *specific* clip. Instance-level pairing thus requires the generator to condition on the source audio.

6 Conclusions, Limitations, and Future Work

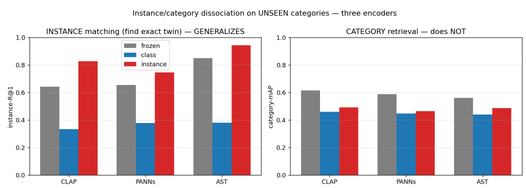
The synthetic-to-real rendering transformation is a learnable, transferable object; category identity is not. **Limitations**: category clustering does not transfer to unseen categories (train per target taxonomy); twins are Stable-Audio audio-init at one operating point per corpus; frozen-embedding heads cannot recover information the backbone discarded; instance pairing requires audio-conditioned generation. SSL encoders (BEATs [Chen et al., 2023], AudioMAE [Huang et al., 2022]) are provided as activatable loaders. **Future work**: domain-relevant corpora (e.g. game and film SFX) and using the sensitive head as a realness metric for generative-audio evaluation.

Reproducibility and release

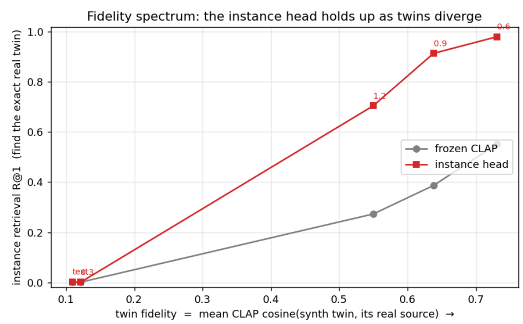
Code (MIT), DCASE and UCS manifests, the CLAP-verification, the generation recipe (event, prompt, seed, steps—bit-reproducible without redistributing restricted audio), the trained heads, and a reusable transform are released. A datasheet [Gebru et al., 2021] and a persistent hosting/maintenance plan are in the appendix.

Table 3: Five-fold leave-classes-out on the UCS corpus: held-out (unseen) categories, mean±std.

variant (unseen categories)	category-mAP	**instance-R@1**	instance-MRR
frozen	0.615 ± 0.052	0.642 ± 0.094	0.731 ± 0.081
class-supcon	0.459 ± 0.072	0.334 ± 0.064	0.439 ± 0.061
instance	0.492 ± 0.047	$\mathbf{0.829 \pm 0.045}$	0.888 ± 0.034



(a)



(b)

Figure 3: (a) The instance/category dissociation on unseen categories, three encoders: instance matching generalizes (left), category retrieval does not (right). (b) Fidelity spectrum: the instance head holds up as twins diverge from their source.

References

Martin Arjovsky, Léon Bottou, Ishaan Gulrajani, and David Lopez-Paz. Invariant risk minimization. *arXiv preprint arXiv:1907.02893*, 2019.

Shai Ben-David, John Blitzer, Koby Crammer, Alex Kulesza, Fernando Pereira, and Jennifer Wortman Vaughan. A theory of learning from different domains. *Machine Learning*, 79:151–175, 2010.

Sanyuan Chen, Yu Wu, Chengyi Wang, Shujie Liu, Daniel Tompkins, Zhuo Chen, and Furu Wei. BEATs: Audio pre-training with acoustic tokenizers. In *ICML*, 2023.

Keunwoo Choi, Jaekwon Im, Laurie Heller, Brian McFee, Keisuke Imoto, Yuki Okamoto, Mathieu Lagrange, and Shinnosuke Takamichi. Foley sound synthesis at the DCASE 2023 challenge. *arXiv preprint arXiv:2304.12521*, 2023.

Zach Evans, Julian D Parker, CJ Carr, Zack Zukowski, Josiah Taylor, and Jordi Pons. Stable audio open. *arXiv preprint arXiv:2407.14358*, 2024.

Eduardo Fonseca, Xavier Favory, Jordi Pons, Frederic Font, and Xavier Serra. FSD50K: An open dataset of human-labeled sound events. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 30:829–852, 2022.

Yaroslav Ganin and Victor Lempitsky. Unsupervised domain adaptation by backpropagation. In *ICML*, 2015.

Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé III, and Kate Crawford. Datasheets for datasets. *Communications of the ACM*, 64(12): 86–92, 2021.

Jort F Gemmeke, Daniel PW Ellis, Dylan Freedman, Aren Jansen, Wade Lawrence, R Channing Moore, Manoj Plakal, and Marvin Ritter. Audio set: An ontology and human-labeled dataset for audio events. In *ICASSP*, 2017.

Yuan Gong, Yu-An Chung, and James Glass. AST: Audio spectrogram transformer. In *Interspeech*, 2021.

Po-Yao Huang, Hu Xu, Juncheng Li, Alexei Baevski, Michael Auli, Wojciech Galuba, Florian Metze, and Christoph Feichtenhofer. Masked autoencoders that listen. In *NeurIPS*, 2022.

Prannay Khosla, Piotr Teterwak, Chen Wang, Aaron Sarna, Yonglong Tian, Phillip Isola, Aaron Maschinot, Ce Liu, and Dilip Krishnan. Supervised contrastive learning. In *NeurIPS*, 2020.

Qiuqiang Kong, Yin Cao, Turab Iqbal, Yuxuan Wang, Wenwu Wang, and Mark D Plumbley. PANNs: Large-scale pretrained audio neural networks for audio pattern recognition. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 28:2880–2894, 2020.

Rolf Nordahl, Stefania Serafin, and Luca Turchet. Real, foley or synthetic? an evaluation of everyday walking sounds. In *Proc. Sound and Music Computing (SMC)*, 2010.

Justin Salamon, Christopher Jacoby, and Juan Pablo Bello. A dataset and taxonomy for urban sound research. In *ACM Multimedia*, 2014.

Baochen Sun and Kate Saenko. Deep CORAL: Correlation alignment for deep domain adaptation. In *ECCV Workshops*, 2016.

Joseph Turian, Jordie Shier, Humair Raj Khan, Bhiksha Raj, Björn W Schuller, Christian J Steinmetz, et al. HEAR: Holistic evaluation of audio representations. In *NeurIPS 2021 Competitions and Demonstrations Track (PMLR vol. 176)*, 2022.

Yusong Wu, Ke Chen, Tianyu Zhang, Yuchen Hui, Taylor Berg-Kirkpatrick, and Shlomo Dubnov. Large-scale contrastive language-audio pretraining with feature fusion and keyword-to-caption augmentation. In *ICASSP*, 2023.

Zhirong Wu, Yuanjun Xiong, Stella X Yu, and Dahua Lin. Unsupervised feature learning via non-parametric instance discrimination. In *CVPR*, 2018.

A Datasheet, hosting, and maintenance

The appendix provides the Gebru et al. [Gebru et al., 2021] datasheet (motivation, composition, collection, preprocessing, uses, distribution), per-class/per-split statistics, the licensing matrix (CC-licensed audio redistributed; BBC and restricted audio shipped as a fetch manifest plus the generation recipe), and the maintenance plan (versioned manifests; the code regenerates all derived artifacts from the manifest and the source archives).

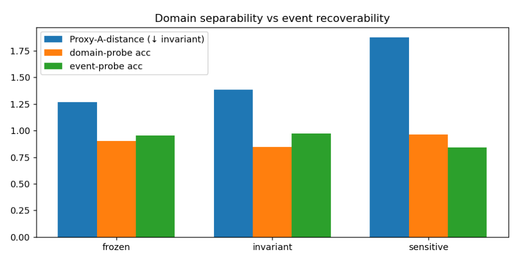
B Corpus statistics and per-event analysis

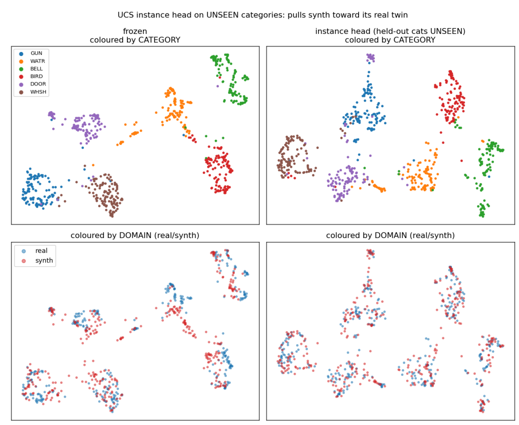
Table 4: Per-domain, per-split clip counts. DCASE-T7 real test is DCASE's source-disjoint eval set; the UCS corpus uses FSD50K's train/val/eval, uploader-keyed.

corpus	domain	train	val	test
DCASE-T7	real	4,150	700	700
DCASE-T7	synth	18,165	3,884	3,851
UCS (34 cls)	real	8,315	999	4,265
UCS (34 cls)	synth (twins)	6,636	719	3,065

Table 5: Per-event frozen-CLAP cross-domain (synth→real) mAP on DCASE-T7. Fast, granular transient/mechanical sounds transfer worst.

	motor	gunshot	dog_bark	sneeze	rain	footstep	keyboard
mAP	0.852	0.810	0.809	0.793	0.752	0.718	0.692





(a) (b)

Figure 4: (a) Proxy-A-distance and event/domain linear-probe accuracy across the DCASE heads. (b) UMAP of the instance head on unseen UCS categories: it mixes real/synth (domain) more than frozen.